

Wrap-up: Ethics, Reproducibility, and Next Steps

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We have become someone who can read, run, and adapt Python code for social research.

What You Have in Your Notebook

- A working Python setup
- Code to open and describe a dataset
- Code to compare groups and run tests
- Code to make four kinds of charts
- A complete analysis pipeline you can reuse

This notebook is a template. Rename it, replace the data file, and you have a starting point for your own research.

Ethics 1: Missing Data Is Not Random

- In our dataset, a few households did not report income
- Who are they? Often the poorest
- Dropping them silently changes the result
- Imputing a median assumes missing households look like the rest
- Neither choice is neutral

Every decision about missing data is a claim about the social world. Justify it in your methods section.

Ethics 2: Reporting Differences Carefully

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Do not write "rural households earn less". Write "rural households earn less because of structural differences in opportunity". The framing is the ethics.

Ethics 3: Honest Charts

- A truncated y-axis exaggerates tiny differences
- Cherry-picking a time period hides trends
- Removing outliers makes the data look cleaner than it is
- Using dramatic colours manipulates emotion
- Good charts show the full picture

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A chart that starts its y-axis at 90 instead of 0 can make a one-point difference look enormous. This is a common form of visual dishonesty.

Reproducibility 1: Making Everything Open

- Every step of our analysis is in the notebook
- Nothing was done in our heads
- Nothing was done in a hidden spreadsheet
- Anyone can read and re-run it
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Reproducibility means: another researcher can take your data and your code and get the same answers. This is the modern standard.

Reproducibility 2: Record Your Tools

```
import pandas as pd
import numpy as np
import scipy

print("pandas:", pd.__version__)
print("numpy:", np.__version__)
print("scipy:", scipy.__version__)
```

Different versions of a tool can give slightly different answers. Always record which versions you used.

Reproducibility 3: A Simple Folder Structure

```
my_study/  
  data/  
    village_survey.csv  
  notebooks/  
    analysis.ipynb  
  figures/  
    income_by_region.png  
  README.md
```

A clear folder structure tells a reader where to look. Write a short README file - five lines is enough.

Where to Share Your Work

- **Zenodo** - free, gives a DOI, good for papers
- **Harvard Dataverse** - used widely in social science
- **OSF** - good for pre-registration and materials
- **GitHub** - version control; pair with Zenodo for a DOI
- **ICSSR Data Service** - Indian social science data

Sharing data and code is now standard practice. Many journals require it before publication.

Common Mistakes to Avoid

- Analysing before looking at the data
- Ignoring missing values
- Reporting only significant results
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- Using small samples for broad claims
- Forgetting to save the package versions

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Careful researchers report everything, including results that did not turn out as expected.

What You Can Learn Next

More statistics

- Regression with controls
- Multilevel models
- Panel data analysis
- Network analysis

More tools

- Text analysis
- Geospatial mapping
- Interactive dashboards
- Version control with Git

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You do not need all of these. Pick one that fits your research question and learn it well.

Recommended Reading

Books

- McKinney, *Python for Data Analysis*
- Downey, *Think Stats* (free online)
- Healy, *Data Visualization*
- D'Ignazio & Klein, *Data Feminism*

Online

- pandas documentation
- seaborn gallery
- scipy.stats reference
- statsmodels notebooks

Read one book, not all four. Depth beats breadth when learning a new tool.

*No statistical method can answer the question:
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That remains the work of interpretation.
Python simply makes the path to that question clearer.

Thank You

The notebook, dataset, and code are yours.

Adapt them to your own research.

Share them with your collaborators.